Table 6.2 Details of the five major mass extinctions since the Cambrian period

Extinction Period	Date (Mya)	Estimated % of Marine Animal Extinction	
		Genera	Species
End of the Ordovician	440	61	85
End of the Devonian	365	55	82
End of the Permian	245	84	96
End of the Triassic	208	50	76
End of the Cretaceous	65	50	76

K-T Extinction refers to the German word Cretaceous and Tertiary periods.

Global extinction eliminates most of the species on a large scale or larger taxonomic groups in the continent or the Earth. Snow ball Earth and extinction following elevation in CO_2 levels are example. Extinction events opens up new habitats and so can facilitate the radiation of organisms that survived the mass extinction.

Summary

Evolutionary Biology is the study of history of life forms on Earth which originated on Earth millions of years ago. How Earth originated, how life originated, what is the place of man in the universe are all general questions. This chapter deals with several theories to explain the life on Earth. Evidence from the fossil record and many other areas of biology like embryology, anatomy and molecular biology indicates a common ancestry.

The theories advanced by Lamarck, Darwin, Hugo de Vries explained the intricate evolutionary process. Geological time scale with different eras, periods and epochs gives an idea about the dominant species in those days. The mathematical distribution of gene and genotype frequencies remains constant in a small population was contributed by Hardy and Weinberg in 1608. Natural Selection and gene pool are the important factors those affect Hardy Weinberg equilibrium.

Human evolution states that humans developed from primates or ape like ancestors. The emergence of Homo sapiens as a distinct species from apes and placental mammals in brain size, eating habit and other behavior proves that 'Ontogeny recapitulates Phylogeny'.

Separation of a single population into its subunits preserves the genetic integrity. Types of isolating mechanism and speciation prevent interbreeding of different species. Various causes of extinction of animals and the levels of impact are dealt with in the lesson.

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Evaluation

1) The first life on earth originated
   a) in air b) on land
   c) in water d) on mountain
2) Who published the book "Origin of species by Natural Selection" in 1859?
   a) Charles Darwin b) Lamarck
   c) Weismann d) Hugo de Vries
3) Which of the following was the contribution of Hugo de Vries?
   a) Theory of mutation
   b) Theory of natural Selection
   c) Theory of inheritance of acquired characters
   d) Germplasm theory
4) The wings of birds and butterflies is an example of
   a) Adaptive radiation
   b) convergent evolution
   c) divergent evolution
   d) variation
5) The phenomenon of " Industrial Melanism" demonstrates
   a) Natural selection
   b) induced mutation
   c) reproductive isolation
   d) geographical isolation
6) Darwin's finches are an excellent example of
   a) connecting links b) seasonal migration
   c) adaptive radiation d) parasitism

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